MINING DEALER BEST PRACTICE

Connection Bypass on Main Cooling for 3500 Engine Test

Maintenance and Repair | Application | Component Life Management | **Component Renewal (CRC)** | MARC Management

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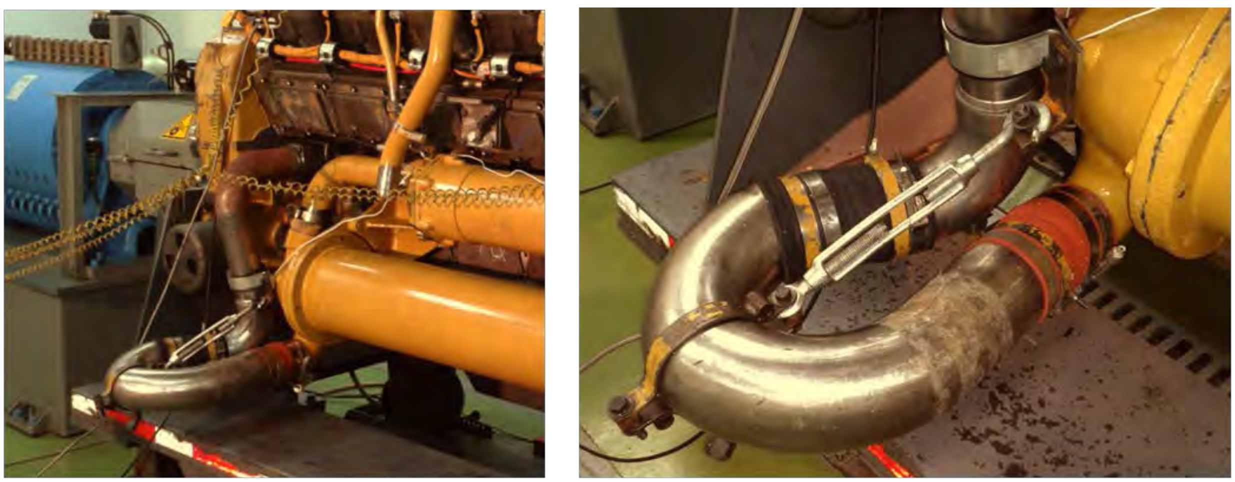
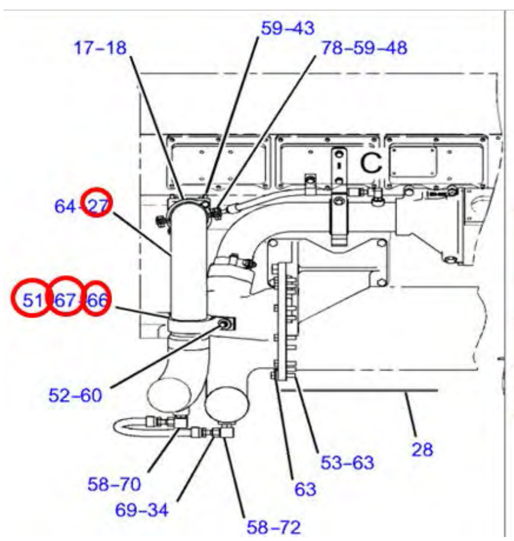
1605-4.03-1434

1.0 Introduction

To Conduct the tests on the dynamometer which prefixes are ATY and FDB (793C / D), the hydraulic oil cooler circuit on the engine block needs to be shut down. Since the SISWEB does not provide an immediate or direct connection between these two points, a horseshoe-shaped tube has been designed to meet this requirement. The problem is that once the device was disconnected a large amount of hot water was leaked during the test creating the risk of burning the components around it and even causing injuries to the technicians. A tension gripping system was designed for the same device in order to hold it more securely, but some problems came up because the vibrations of engine. (See Fig. 1.)





793D Off-Highway Truck, Powered By 3516B Engine
Media Number - SEBP3976-134 I Publication Date - 01/04/2016 I Date Updated - 25/04/2016

***235-0287* LINES GP-WATER -LOWER**
S/N 8WM979-1691
PART OF 235-0300, 288-4348 ENGINE AR
ALSO AN ATTACHMENT

27		NPR	1	287-3719	1	TUBE AS
51	2	NPR	1	103-0892	2	GROMMET (CCR Part)
66		NPR	1	6V-6782	2	CLIP (SLOT)
67		NPR	1	6V-6783	2	CLIP (TAB)

Fig. 1.

This Best Practice aims to design a main cooling system connecting device during the 3500 engines test in order to solve the aforementioned problems.

2.0 Best Practice Description

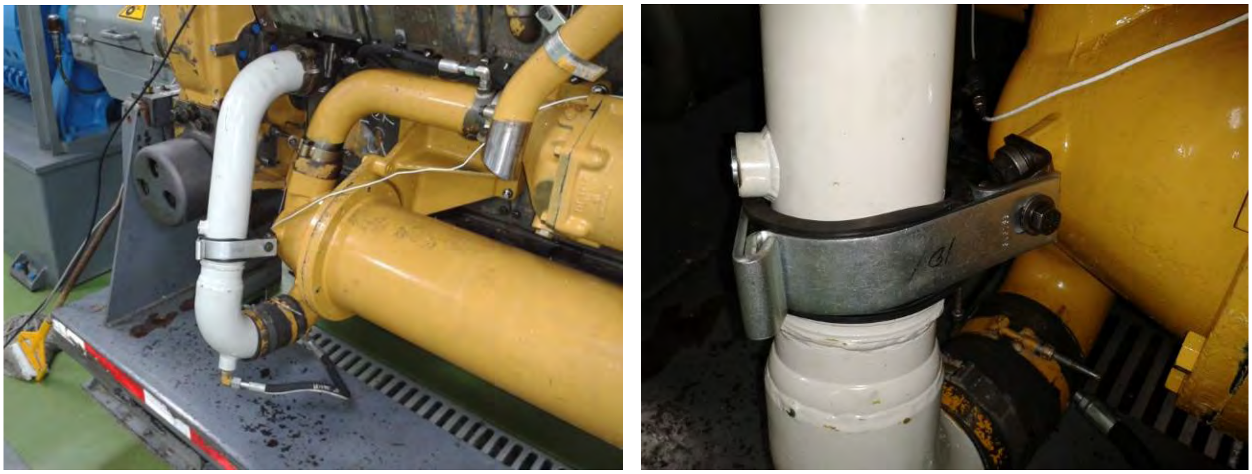


Fig. 2.

A direct link between the cylinder block and the hco cooler with a C-shaped device and a bedpan CAT tube (340-1590) is designed, which has the same dimensions of the inlet cooler. The tube is 4.5 inch diameter and 32 inch long. It has two soft angles that allow the free passage of coolant without creating turbulences. The original material is factory galvanized steel. Its structure is made up of two welded pieces together that gives a C-shaped appearance. The tube is secured with two clamps and an original factory grommet (6V-6782, 6V-6783, 103-0892). (See Fig. 2).

3.0 Implementation Steps

To implement the improvement the 340-1590 tube is taken and cut thru the lower outlet of the coolant and welded back again at the same point but in the opposite direction to its original design to match the coolant inlet. Then the two inlets are put together by means of a single hose and special clamps are then attached. To hold it safely, the two clamps and the grommet are installed (PN: 6V-6782, 6V-6783, 103-0892) into the ½ " diameter thread that the oil cooler elbow has.

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4.0 Benefits

1) It prevents the device to be ejected unexpectedly due to the force of the hot coolant at full engine load by using fewer joint points with hoses and to have a better grip to CAT original accessories.

2) It improves the installation times of accessories during the readiness of the 3516 engines in the dynamometer.

3) Less parts resources for the device coupling to the engine.

5.0 Resources Required

All the resources used must be identified for the design, fabrication, commissioning (implementation) and thus maintainability of the improvement, such as:

- Personnel: Cutting and machining and welding.
- Total investment: ~$650 (Parts and Labor).
- Appoint manufacturing plans: (pending by the supplier)
- Equipment and tools: electric hacksaw cutter, TIG welding equipment.

6.0 Supporting Attachments / References

None

7.0 Related Best Practices

None

8.0 Acknowledgements

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